Fiscal Year:	FY 2011	**Task Last Updated:**	FY 10/05/2010
PI Name:	Cromwell, Ronita Ph.D.		
Project Title:	Bed Rest Standard Measures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2004	**End Date:**	12/01/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	October 2010: Buccello-Stout, Regina (Universities Space Research Association) Removed from protocol. Zanello, Susana (Universities Space Research Association) Added to protocol.		
COI Name (Institution):	Ware, David (University of Texas Medical Branch) Yarbough, Patrice (Universities Space Research Association) Zanello, Susana (Universities Space Research Association)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Flight Analogs Project Plan outlines a multi-phase approach for implementation of NASA-funded bed rest studies. This study is currently conducted at the NASA Flight Analogs Research Unit located at The University of Texas Medical Branch (UTMB) in Galveston, Texas. Although head-down bed rest has been used as a spaceflight analog to study changes in physiologic function associated with spaceflight for many years at many different institutions, standards for conducting such studies have not previously been established. Conditions of bed rest have been investigation or site specific. Most studies have had no direct comparisons of bed rest data with spaceflight data. As a consequence, systematic, multi-disciplinary comparisons across studies and comparisons of bed rest and actual spaceflight data have not been possible. To address these issues, the FAP has developed standard conditions for implementation of all NASA-funded bed rest studies and a set of standard measurements to be collected from all bed rest subjects. The standard measurements replicate, for the most part, measurements obtained on astronauts before and after long duration missions. Spaceflight and bed rest measurements are taken by the Johnson Space Center Laboratories and are used for three purposes: they make possible a direct comparison between responses to spaceflight and responses to head-down bed rest; they provide a characterization of human responses to long duration head-down bed rest; and they provide the ability to assess candidate countermeasures in a multi-disciplinary manner by evaluating the effects on the targeted systems as well as possible unexpected effects on non-targeted systems. The following measures are included in the standard measures complement. Nutritional status and immune function are assessed through blood, urine and saliva samples. Standard clinical laboratory assessment of blood and urine are used to monitor subjects' health status throughout the study. Dual Energy X-Ray Absorptiometry (DXA) in combination with bone biomarkers are used to assess changes in bone. Physical Fitness is assessed through measures of muscle strength, endurance, flexibility and maximum aerobic capacity. Posturography and monosynaptic T-reflex testing provides assessments of neurological function. The cardiovascular system is assessed through measures of orthostatic tolerance, blood volume and cardiac function. These multidisciplinary measures are provided to investigators to enhance interpretation of system-specific results. By standardizing conditions and measures for all bed rest studies, NASA is able to compare results across bed rest studies in a multidisciplinary manner. The standard set of measures will be collected on as many bed rest subjects as is feasible to provide a multi-discipline evaluation of potential countermeasures. These measures will be integrated with the peer-reviewed science requirements on a non-interference basis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In FY2010, standard measures were used in 2 bed rest campaigns. The first was the Campaign 3 60-day, long-duration study. The second was a short-duration, 30-day bed rest complement. Included in this report is a summary of activities from these studies for FY2010. Campaign 3: The Campaign 3 study entitled "Gender Differences in Bed Rest: Autonomic, Neuroendocrine, and Vascular Responses in the Lower and Upper Extremities" was an ongoing study that completed operations in 2010. A total of 23 healthy subjects (10 females and 13 males) completed this study. The objective of this study was to determine which cardiovascular attributes contributed to the difference in incidence of orthostatic intolerance between genders. Post spaceflight orthostatic intolerance, the inability to maintain adequate blood pressure while in an upright position, remains a significant concern to NASA. Orthostatic intolerance is especially problematic for long-duration space missions. Female astronauts exhibit greater orthostatic intolerance than their male counterparts. The mechanisms behind this problem were studied in the ground-based analog of head-down tilt bed rest. Six degrees head-down tilt bed rest induced cardiovascular changes similar to that observed in spaceflight. During the study, participants were admitted to the NASA Flight Analogs Research Unit located at the University of Texas Medical Branch in Galveston, Texas. In the first phase of the study, participants were free to move about the bed rest facility. Baseline testing was completed during this phase. In the next phase, long-duration bed rest, participants were placed in bed and tilted in a 6° head-down tilt position for 60 or 90 days. Participants remained in this position for all activities including testing and hygiene. For the final phase of the study, participants were able to move freely about the unit while they received rehabilitation and post study testing. The complete battery of standard measures was collected on these participants. Selected results are summarized for this report. Measures of body fluids during each day of bed rest were less than that reported during pre-bed rest baseline measures ($p < 0.05$). Results of body fluid shifts indicated that due to body size, females exhibited lower values than males. However, when adjusted for body surface area, both males and females showed similar decreases in body fluids during bed rest. Body fluids recovered to pre-bed rest values by day 4 after bed rest. Results of the tilt test for orthostatic tolerance indicated that subjects had a lower probability of completing the tilt test on the day they got up from bed as compared to pre-bed rest values. Additionally, the probability of completing the tilt test 4 days after getting up from bed is not different from that of pre-bed rest values. Tilt test results for males and females were only different prior to bed rest ($p=0.038$) and did not demonstrate differences for post bed rest measures. Relative peak oxygen uptake (VO_2pk) values that represent cardiorespiratory fitness levels normalized to body weight were lower at post bed rest when compared to pre-bed rest values ($p<0.05$). There were no gender differences in relative VO_2pk at baseline and immediate post bed rest. However, at 12 days post bed rest, VO_2pk was higher in men than women ($p<0.05$). This finding suggests that men recover more quickly from long duration bed rest than women. Results of the isokinetic muscle strength and endurance parameters demonstrated that for eleven of the twelve parameters measured, significant decreases were observed ($p = 0.05$). While differences in gender were observed, these differences were due to males' greater overall strength and not a gender response to bed rest. 30-day Bed Rest Complement: This shorter duration bed rest study incorporated investigations of 4 scientists. By integrating the investigations on a non-interference basis, the same subject sample was used by all investigators in an effort to more efficiently complete the research. A total of 12 healthy participants completed 30 days of 6° head-down tilt bed rest for this study. Upon admission to the NASA Flight Analogs Research Unit, participants completed 2 weeks of pre-bed rest baseline testing. This was followed by the 30-day head-down tilt bed rest period. Following the bed rest phase, participants spent 2 weeks in post bed rest testing and rehabilitation. Investigator studies in this complement included a study entitled "Rapid Measurements of bone loss using Tracer-less

	Calcium Isotope Analysis of Blood and Urine." In this study, calcium isotope composition in blood and urine samples was used to develop methods for early detection of changes in bone mineral balance. A second study integrated into this complement was "Validation of Near Infrared Spectroscopy (NIRS) Measures following Bed Rest." For this study, the NIRS device was validated as a non-invasive method to continuously measure oxygen consumption during cycling exercise to maximal effort. During these same cycling exercise sessions, participants also took part in the study "Non-invasive Device for Measuring Core Temperature during Maximal Exercise." This study, examined a non-invasive method using a double sensor placed on the head and chest to assess core temperature of the body during exercise. A forth study in this campaign was "Methods for the Assessment of Gastrointestinal Physiology and Function in a Reduced Gravity Analog." Gastrointestinal function was assessed using SmartPill® technology. Participants ingested a capsule containing sensors that monitored pH, temperature, pressure and regional transit time throughout the gastrointestinal tract. Standard measures were integrated with these investigator studies. A subset of the standard measures battery was used to provide a multidisciplinary assessment of each participant's physiological response to 30-days of bed rest. A subset of measures was used because 30-days of bed rest were not long enough to observe changes in some physiological systems. The subset of measures included: cycle ergometry, isokinetic testing, functional fitness, nutrition assessment, posturography and T-reflex testing. Study operations were completed and data analyses are ongoing.
Bibliography Type:	Description: (Last Updated: 05/16/2019)
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